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Beacon Outfitters

Retail replenishment | Decision brief

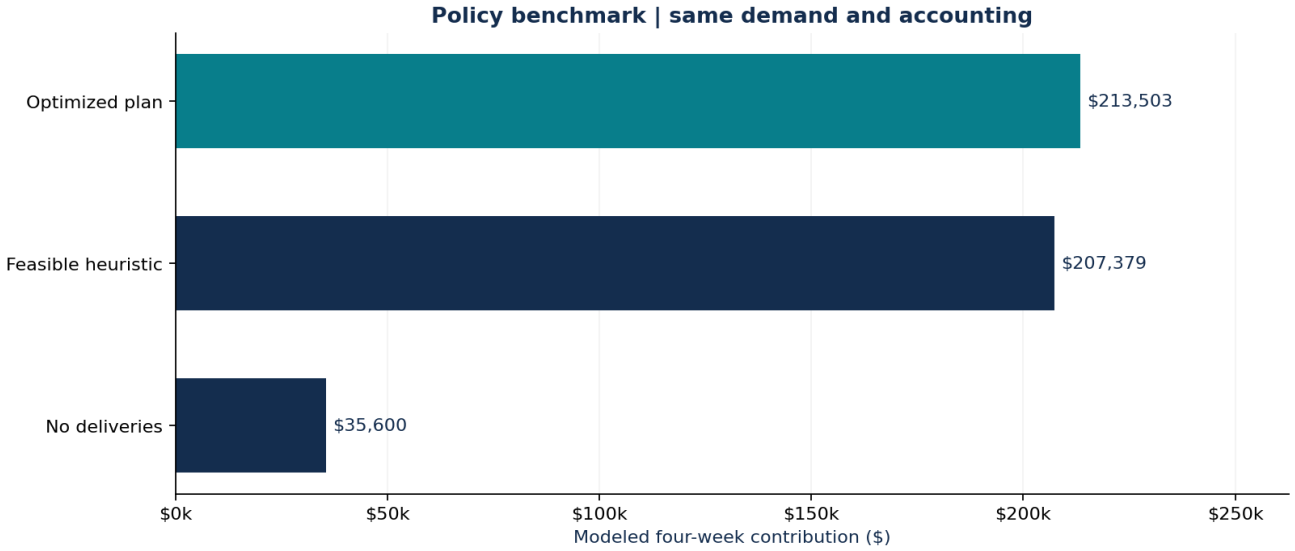
Plan 512 cases (4,028 units) on 27 visits across 12 stores and four weeks. The verified benchmark yields $213,503.10 in modeled contribution, $6,123.90 (2.95%) above a feasible forecast-driven heuristic.

Decision: allocate scarce supply, choose delivery weeks and balance sales margin against stock holding and truck costs. These are benchmark outcomes, not realized company savings.

Decision metric	Verified result
Modeled contribution	$213,503.10
Heuristic uplift	$6,123.90 / 2.95%
Demand fill rate	96.13%
Best +100-unit opportunity	J200 parkas: $10,169.20
Forced alternate-week loss	$3,124.00

Recommended action

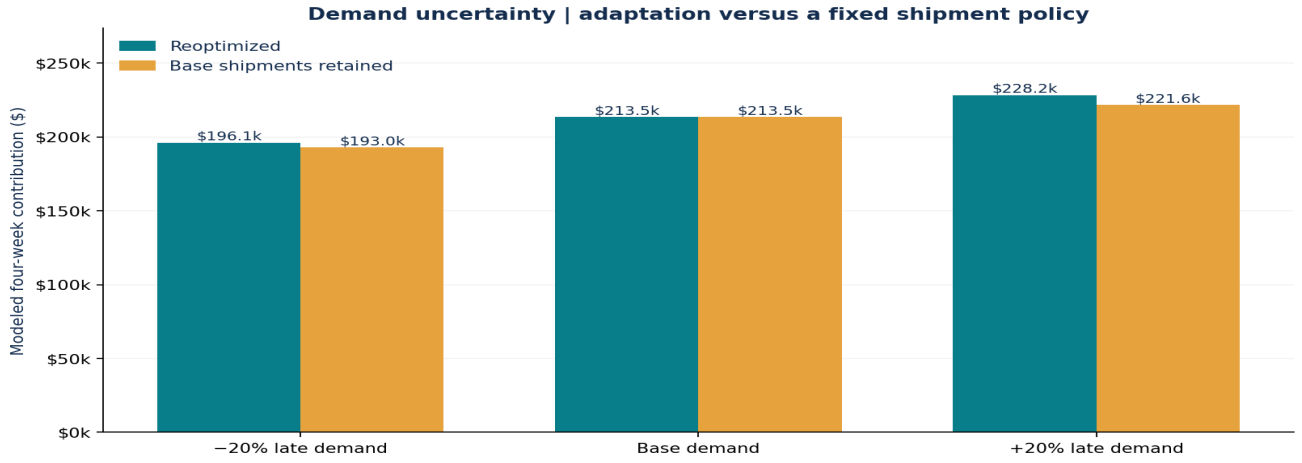
Use the store-specific delivery calendar rather than a blanket alternating-week rule. Evaluate 100 additional J200 parkas released in week 1: their gross incremental contribution is $10,169.20 before any new procurement premium. Refresh the allocation when demand or releases change.



Uncertainty and execution

Weeks 3–4 demand	Reoptimized contribution	Base shipments retained
−20%	$196,074.61	$192,979.83
Base	$213,503.10	$213,503.10
+20%	$228,223.15	$221,594.99

The stress tests change all eight SKUs in weeks 3–4 and retain exact fractional demand. Reoptimization is perfect-foresight adaptation; fixed-policy evaluation allows sales and remaining inventory to adjust while shipments and visits stay fixed. Neither is a forecast confidence interval.



Store-specific delivery schedule

Store	Visit weeks	Units
S01	1, 2, 4	562
S02	1, 2, 4	406
S03	1, 2, 4	370
S04	1, 2, 4	330
S05	1, 3	376
S06	1, 3	334
S07	1, 4	280
S08	1, 4	248
S09	1, 3	326
S10	1, 3	292
S11	1, 3	282
S12	2	222

Why the result is credible

Contribution reconciles to $221,013.00 sales margin less $1,399.90 holding and $6,110.00 visits. Gurobi and CP-SAT independently prove the same base objective. Every extracted plan is checked for flow balance, demand, shared shelf limits, truck activation, cumulative supply and whole packs.

Data and model

Five validated CSVs describe 12 stores, 8 SKUs and 4 weeks. The formulation has 1,584 variables with integer cases and binary visits. Actual truck capacity supplies the linking bound. The pipeline records input/model hashes, runtime versions, solver bounds, gaps and independent verification evidence. Provided demand forecasts are inputs, not predictions trained here.

Scope and limitations

This is a deterministic retail benchmark. Opening-stock sales follow the margin convention; unit costs are counted once. No routing, supplier lead times, forecast learning, terminal salvage or operational service floor is assumed. Time limits and tied allocations can affect solutions on different hardware. Deployment needs rolling validation, calibrated uncertainty and operational review.

Contributors and assistance

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ChatGPT assisted debugging and technical explanations. OpenAI Codex assisted implementation, automated tests, documentation and artifact generation. Numerical claims are solver-derived and independently checked; contributors remain responsible for interpretation.

Source: Hasan Arslan’s public benchmark data and modeling reference; full credit, original revision and hashes are in docs/data_provenance.md.